Introduction to Molecular Dynamics

Laboratory for Chemistry and Life Science, Institute of Integrated Research, Institute of Science Tokyo
Tateyama–Ando Lab. Ryoma Sasaki

1. Flow of MD simulation
2. Energy calculation
3. Time integration
4. Statistical mechanics; T & P control
5. Data analysis

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Solving ma = F numerically

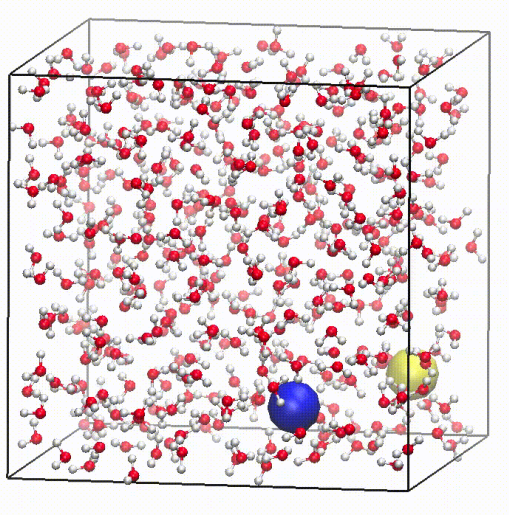
$$ma = F = -\frac{\partial U(r)}{\partial r}$$

Theory questions

- What kinds of force (energy) calculations are there?
- No analytical solution
  → How to solve numerically?
- How to link micro and macro?
  → **Statistical mechanics**
- How does Gaussian differ?

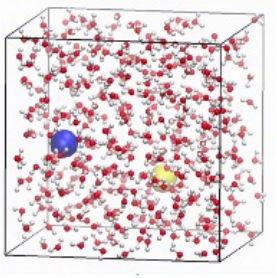
Software/hardware questions

- Which software to use?
- Environment setup
- Linux
- Data analysis
- Parallelism, CPU/GPU



MD simulation workflow

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F = -\frac{\partial U(r)}{\partial r}$$



①Place atoms, set velocities

②Compute forces

$$F = -\frac{\partial U(r)}{\partial r}$$

Repeat

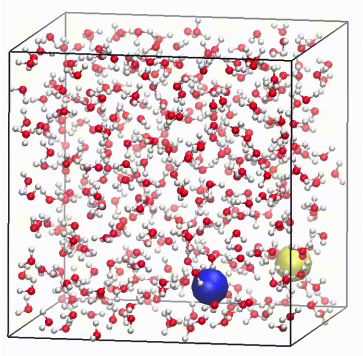
③Move atoms

$$r(t+\Delta t) = r(t) + v(t)\Delta t + \frac{1}{2}\frac{F(t)}{m}\Delta t^2 + \mathcal{O}((\Delta t)^3)$$

$$v(t+\Delta t) = v(t) + \frac{\Delta t}{2}\left[\frac{F(t)}{m} + \frac{F(t+\Delta t)}{m}\right] + \mathcal{O}((\Delta t)^3)$$

Δt later: positions

Trajectory



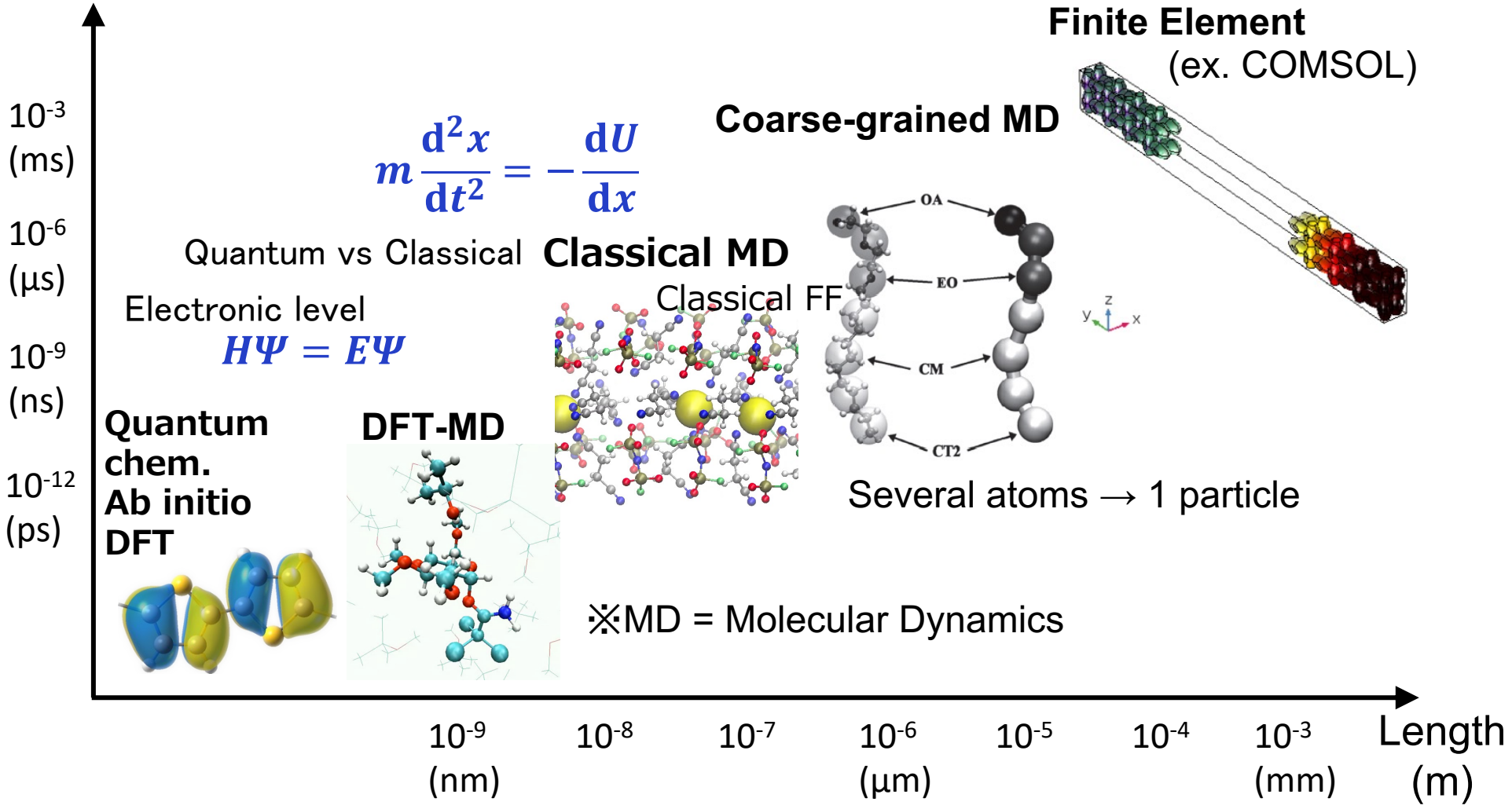
④Sampling, statistics → structure & properties

Total cost = Energy calc. x No. of samples

∝ energy accuracy
length scale (system size)

∝ statistical accuracy
time scale

Time and length scales of molecular simulation



How accurately to treat the energy (*F*) calculation?

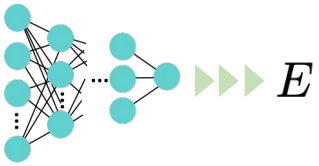
Trade-off between accuracy and scale

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Types of energy calculation

	Classical FF	MLFF	DFT
	$E(\boldsymbol{r}) = K_{\mathrm{bond}}(r-r_0)^2 + K_{\mathrm{angle}}(\theta-\theta_0)^2 + 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + \cdots$		$\hat{H}\Psi = E\Psi$
Accuracy	Low	**Mid–high** ≒DFT	**High**
Cost	**Low**	Fairly low	High
Required know-how	High	Low–mid	**Low**
Software (MD runs)	• LAMMPS • AMBER • GROMACS • OpenMM	• **ASE** (• LAMMPS)	VASP CP2K QuantumEspresso
Analysis	←	**ASE** **(Atomic Simulation Environment)**	→

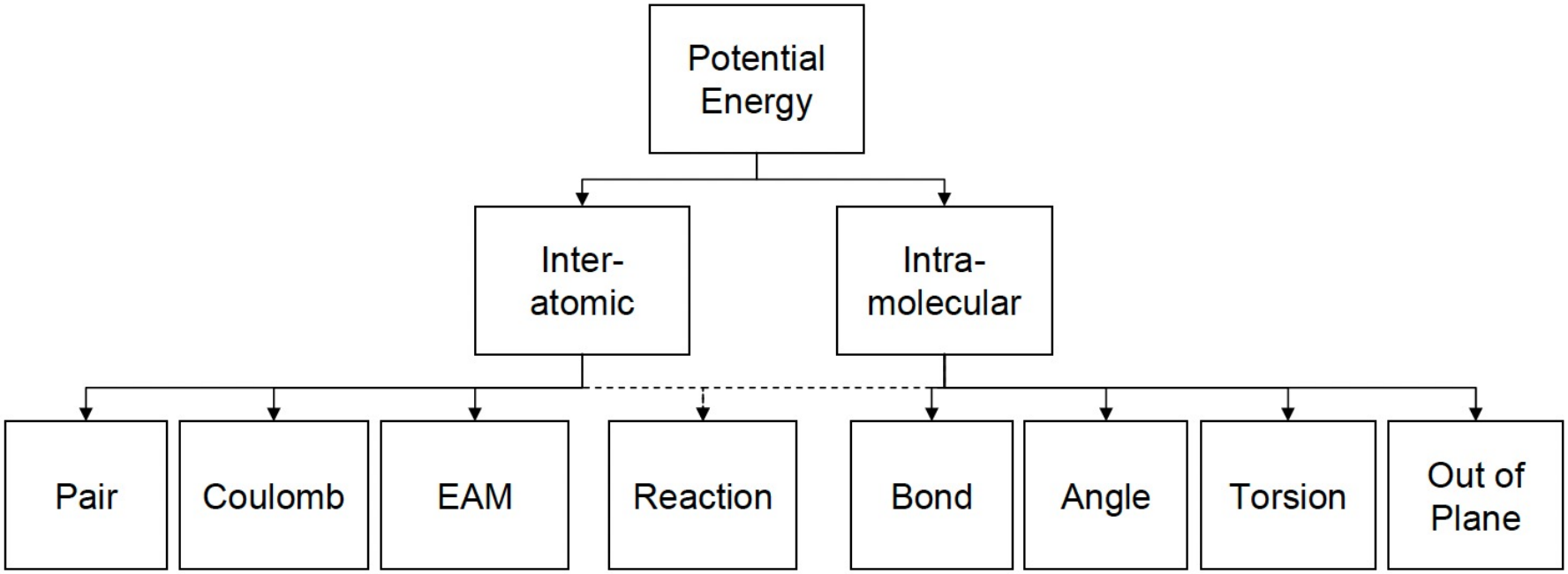
- **With ASE, you can run calculations at every level**
- MLFF balances know-how & accuracy: ideal to learn

Types of energy calculation

	Classical FF	MLFF	DFT
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Classical force field

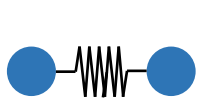
Force field / empirical parameters per atom type & interaction

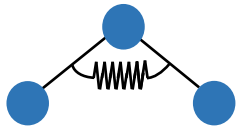


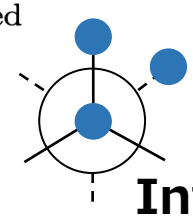
Classical force field

■ Organic molecular systems

$$E(\{\boldsymbol{r}\}_{i=1,2,\ldots,N}) = \sum_{\text{bonds}} K_{\text{bond},i}(r - r_0)^2 + \sum_{\text{angles}} K_{\text{angle},i}(\theta - \theta_0)^2 + \sum_{\text{dihed}} K_{\text{dihed},i}\,[1 + \cos(n\phi + \delta)]$$







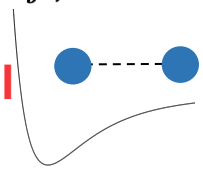
Intramolecular

$$+ \sum_{i,j} \epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right] + \sum_{i,j} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \quad +\ldots$$

Non-bonded

Lennard-Jones potential

(van der Waals force)





Representative force fields

- Non-polarizable models
  - –Class I: GAFF, OPLS, Dreiding, CHARMM, CVFF
  - –Class II (cubic/quartic terms, cross terms): COMPASS, CFF, PCFF
- Polarizable models (Class III): Drude, AMOEBA, APPLE&P
- Chemical reactions: ReaxFF

Lennard–Jones Potential

$$\sum_{i,j} \epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right]$$

Originally used for rare-gas atoms, but works well for organics too

Repulsion Attraction

- Short-range
- Pauli repulsion

- Long-range
- Dispersion

← r^{-6} **dependence can be derived theoretically!**

JCTC Journal of Chemical Theory and Computation

2024: Lennard-Jones potential turns 100!

pubs.acs.org/JCTC Review

100 Years of the Lennard-Jones Potential

Peter Schwerdtfeger* and David J. Wales*

Cite This: *J. Chem. Theory Comput.* 2024, 20, 3379–3405 Read Online

First liquid MD simulation; A. Rahman, *Phys. Rev.* **136**, A405 (1964).
In the exercise, we simulate liquid argon

Classical force field

■ Inorganic systems

- BMH (Born-Mayer-Huggins) potential
  - Ionic crystals

$$E(\{\boldsymbol{r}\}_{i=1,2,\ldots,N}) = \sum_{i,j}\left[A_{ij}\exp\left[(\sigma_i+\sigma_j-r_{ij})/\rho\right] + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} - \frac{C_{ij}}{r_{ij}^6} - \frac{D_{ij}}{r_{ij}^8}\right]$$

- EAM (Embedded Atom Model) potential
  - Metals (ion cores immersed in delocalized electrons)

$$E(\{\boldsymbol{r}\}_{i=1,2,\ldots,N}) = \sum_i F_i(\rho_i) + \frac{1}{2}\sum_{i,j}\phi_{ij}(r_{ij})$$

Energy to embed atom *i* in ρ_i (local electron density)

Pair potential (screened Coulomb)

- SW (Stillinger-Weber) potential, Tersoff potential
  - Covalent materials (e.g. Si, Ge, silica, etc.)
    2-body plus many-body terms

Types of energy calculation

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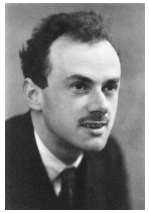
Density functional theory (DFT)

Schrödinger equation

$$H\Psi = E\Psi \qquad H = -\sum_i^n \frac{\nabla_i^2}{2} - \sum_{\alpha,i} \frac{1}{r_{\alpha i}} + \sum_{i<j} \frac{1}{r_{ij}}$$

Kinetic energy | Nucleus–e | e–e

$\Psi(\boldsymbol{r_1}, \ldots \boldsymbol{r_n})$ Electron wavefunction



Dirac

Many-body (N>2) equations: no analytic solution

"The underlying physical laws necessary for ... the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble."

Density functional theory (DFT)

Hohenberg-Kohn theorem: ground-state energy is a **functional of density ρ**

$\rightarrow E[\rho]$

+variational principle $E[\rho_{true}] \leq E[\rho_{trial}]$

[Issues]

- Functional parameters are fitted to experimental data
- Poor description of long-range interactions (e.g. dispersion)
- Density-based theory (individual electrons hard to distinguish)

  $\rightarrow$ spurious interaction of an electron with itself (self-interaction)

Density functional theory (DFT)

Accuracy of functionals

Accurate

GGA: **PBE**, PBEsol, BLYP,...

* Poor for localized, high-l orbitals (e.g. d)
keyword: DFT+U

Standard for DFT-MD

meta-GGA: Scan, **r2Scan**,...

Increasingly common even in DFT-MD

Hybrid Functional: **HSE**, PBE0, **B3LYP**,...

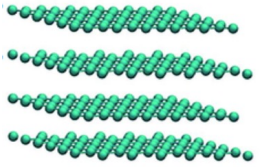
Range-separated: **ωB97X**

Rarely used in DFT-MD

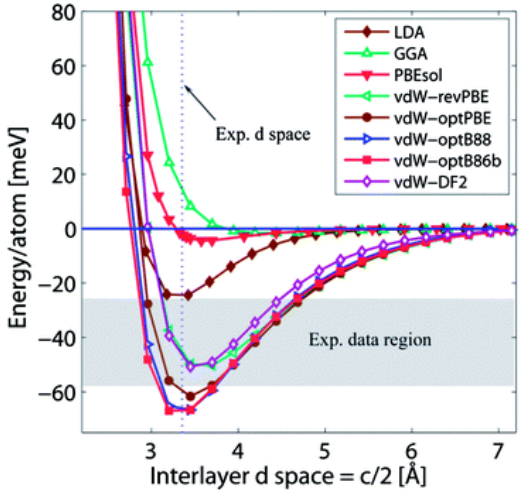
+ Dispersion correction

DFT is poor at long-range interactions
(e.g. dispersion, π-π stacking)

Use e.g. Grimme-D3, especially for organics

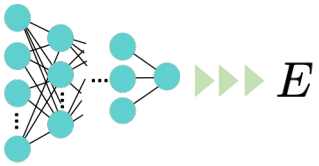


Graphite



RSC Advances, **4**, 4069 (2014).

Types of energy calculation

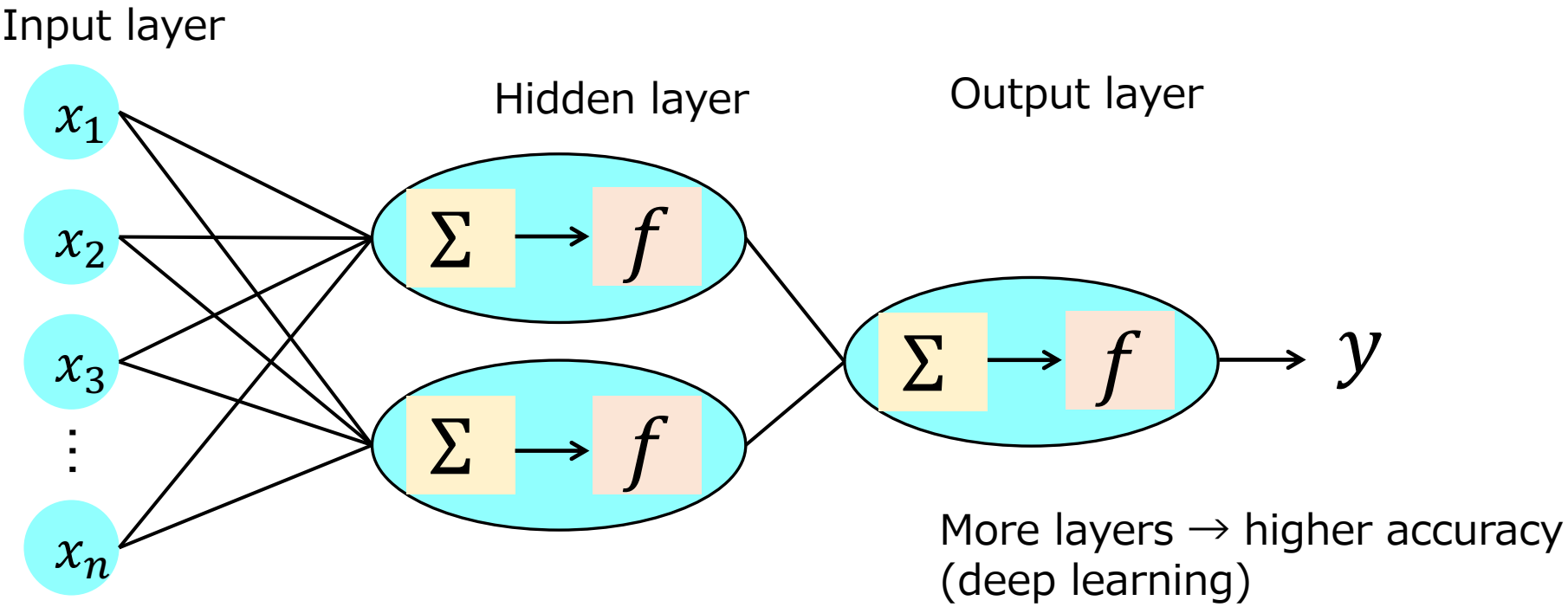
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Machine-learning force field (MLFF)

- Classical FF: limited expressiveness (accuracy)…
  - fixed functional form; reactions, polarization, orbital interactions
- DFT: far too time-consuming…
  - only a few 100 atoms, <1ns

→ More **flexible** than classical FF + **faster** than DFT

Use neural networks (NN) (trained on DFT data → DFT accuracy)



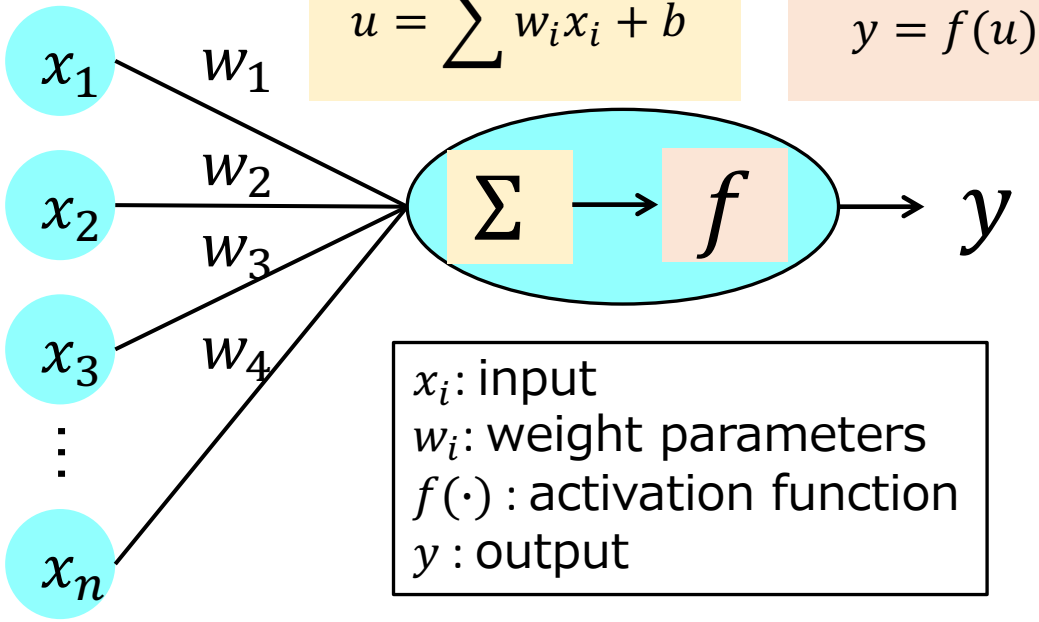
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Descriptor



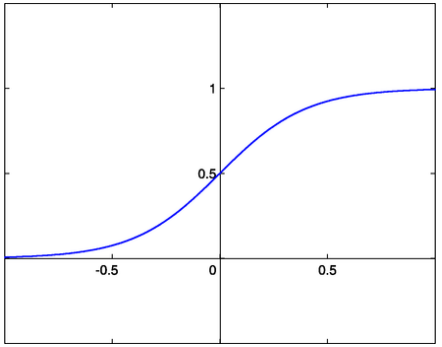
$$u = \sum w_i x_i + b$$

$$y = f(u)$$

x_i: input
w_i: weight parameters
$f(\cdot)$: activation function
y : output

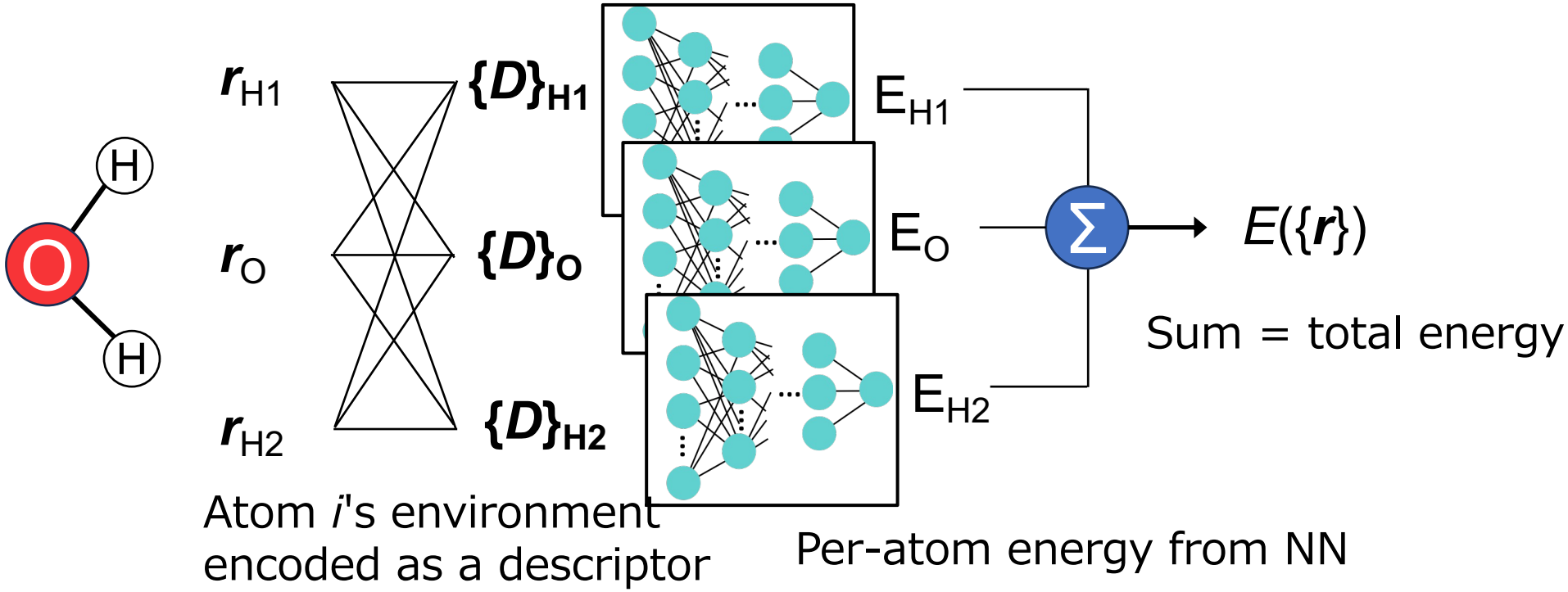
Activation adds nonlinearity (e.g. sigmoid function)

$$\varsigma_a(x) = \frac{1}{1 + e^{-ax}}$$



Machine-learning force field (MLFF)

Neural network potential



Above: descriptor-based neural network potential
Graph neural network potentials also exist

Computational cost: classical FF vs MLFF vs DFT

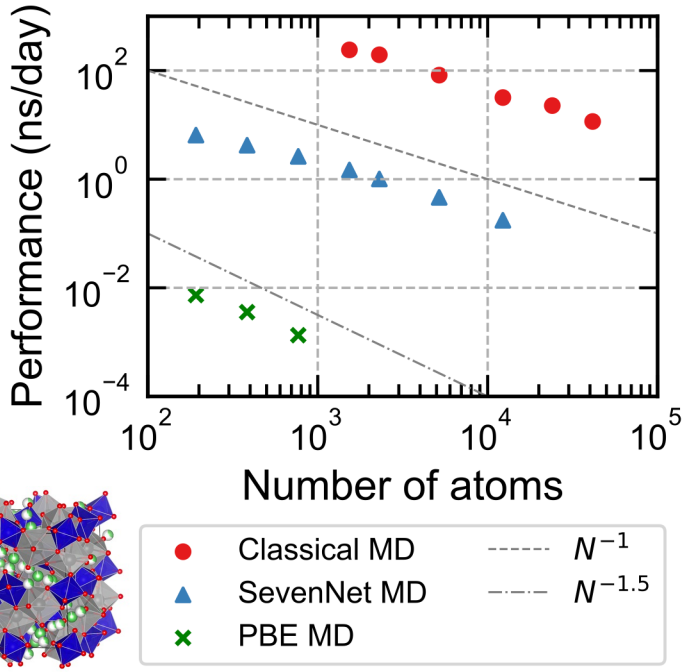
Matbench

Model	MAE ↓	RMSE ↓	R^2 ↑	Training Set	Model Params
eqV2 S DeNS	0.036	0.085	0.788	146K (1.6M) (MPtrj)	31.2M
ORB MPtrj	0.045	0.091	0.756	146K (1.6M) (MPtrj)	25.2M
SevenNet	0.048	0.092	0.750	146K (1.6M) (MPtrj)	842.4K
MACE	0.057	0.101	0.697	146K (1.6M) (MPtrj)	4.7M
CHGNet	0.063	0.103	0.689	146K (1.6M) (MPtrj)	412.5K
M3GNet	0.075	0.118	0.585	63K (188.3K) (MPF)	227.5K
ALIGNN	0.093	0.154	0.297	155K (MP 2022)	4.0M
MEGNet	0.130	0.206	-0.248	133K (MP Graphs)	167.8K
CGCNN	0.138	0.233	-0.603	155K (MP 2022)	128.4K (N=10)
CGCNN+P	0.113	0.182	0.019	155K (MP 2022)	128.4K (N=10)
Wrenformer	0.110	0.186	-0.018	155K (MP 2022)	5.2M (N=10)
BOWSR	0.118	0.167	0.151	133K (MP Graphs)	167.8K
Voronoi RF	0.148	0.212	-0.329	155K (MP 2022)	26.2M
Dummy	0.124	0.184	0.000		

https://matbench-discovery.materialsproject.org/

Parameters: few + multi-GPU parallel

Maximum Performances of
cpu_80 queue (80-cores CPU, 18.75JPY/hour) and
node_q (H100 GPU and 48-cores CPU, 15.625JPY/hour)



($Li_7La_3Zr_2O_{12}$, Class-I Force field)

Speed x1/100– x1/100 x1/10

Classical FF → GNN-based MLFF → GGA → Hybrid GGA

SevenNet: https://github.com/MDIL-SNU/SevenNet/tree/main/sevenn/pretrained_potentials/SevenNet_0__11Jul2024

MLFF: challenges and current status

Issue ①: being "machine learning", works properly only within the DFT training data range
(regions far outside the training domain are hard to handle)
– Run DFT-MD at various temperatures/pressures for diverse data
Combine with active learning etc...

Issue ②: never cheaper than classical MD;
hard to apply where even classical MD is barely feasible

Status: everyone builds their own MLFF for their system
→ Trend: big players build **universal MLFFs (foundation models)**

Industry

MATLANTIS

GNoME

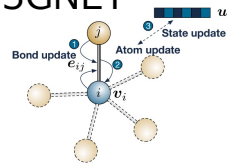
Meta FAIR Chemistry

MatterSim Microsoft

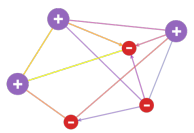
Academia

SevenNet

M3GNET



CHGNet



Will per-system fine-tuning become the norm?

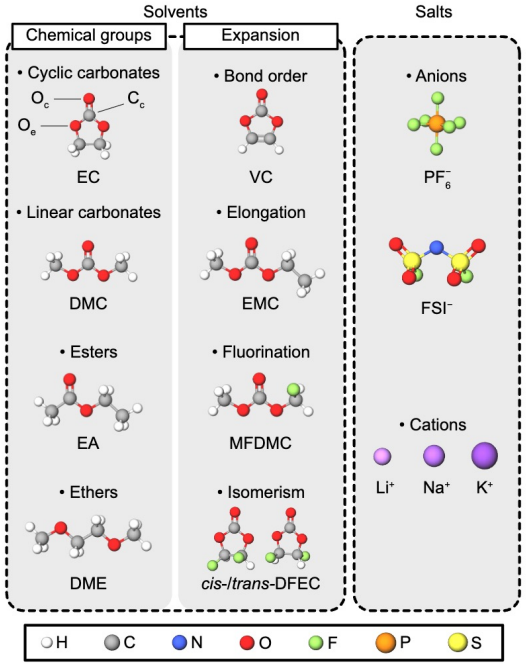
c.f.
Computer Vision, NLP

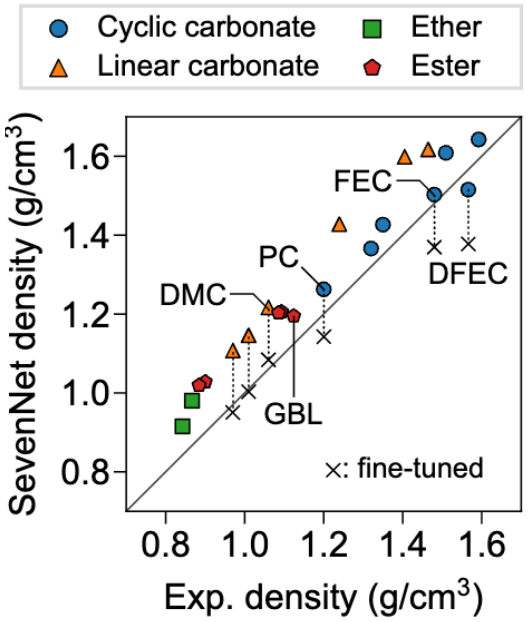
Latest MLFF info: Matbench Discovery dashboard
https://matbench-discovery.materialsproject.org/

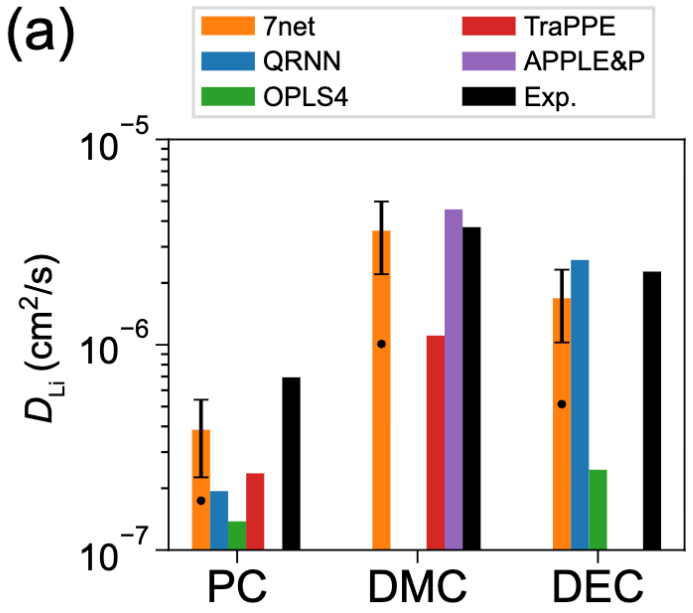
Aside: extrapolating from inorganic to organic

Many universal MLFFs are trained on
Materials Project-based **inorganic databases**
→ **Do they work for organics, e.g. organic electrolytes**?

Study with SevenNet-0 (shown earlier): https://arxiv.org/abs/2501.05211







Surprisingly (?), properties are predicted quite accurately
(though electrolyte decomposition reactions would likely fail⋯)

Aside: Physics Nobel Prize & molecular simulation

◎ Backpropagation

Training a neural network means...

For a neural net $f(\{\boldsymbol{x}\}; w_1, w_2, \ldots)$,

Find parameters that minimize $|y_i - f(x_1, x_2, \ldots, x_n)|^2$
(deviation from true values {yi})
⇨ Need derivatives of the error w.r.t. $w_1, w_2, \ldots$



2024
Physics Nobel Prize

Enabled "automatic" derivatives of complex functions (autodiff)

Autodiff → molecular simulation benefits too!

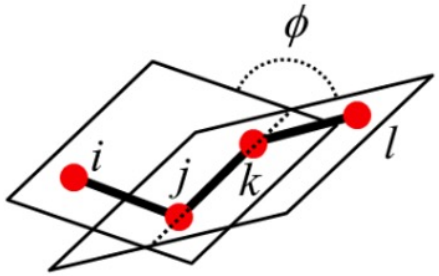
Force $F_{\gamma\alpha} = -\dfrac{\partial E}{\partial R_{\gamma\alpha}}$ (Virial) pressure $\sigma_{\alpha\beta} = -\dfrac{1}{\Omega}\dfrac{\partial E}{\partial \varepsilon_{\alpha\beta}}$

AD-oriented MD code → JAX-MD

(Simple example: https://github.com/aiqm/torchani/blob/master/torchani/ase.py)

Aside: simple energy, yet complex forces?

$$K_{\mathrm{dihed},i}\left[1+\cos(n\phi+\delta)\right]$$



$$\phi = -\mathrm{sign}\left\{\cos^{-1}\left(\frac{\boldsymbol{n}_j\cdot\boldsymbol{n}_k}{n_j n_k}\right), \boldsymbol{r}_{jk}\cdot(\boldsymbol{n}_j\times\boldsymbol{n}_k)\right\}$$

$$\begin{aligned}
\boldsymbol{n}_j &= \boldsymbol{r}_{ij}\times\boldsymbol{r}_{kj}\\
\boldsymbol{n}_k &= \boldsymbol{r}_{jk}\times\boldsymbol{r}_{lk}\\
\boldsymbol{r}_{ij} &= \boldsymbol{r}_i-\boldsymbol{r}_j=-\boldsymbol{r}_{ji}\\
\boldsymbol{r}_{jk} &= \boldsymbol{r}_j-\boldsymbol{r}_k=-\boldsymbol{r}_{kj}\\
\boldsymbol{r}_{lk} &= \boldsymbol{r}_l-\boldsymbol{r}_k=-\boldsymbol{r}_{kl}
\end{aligned}$$

$$\boldsymbol{F}_i^{\mathrm{dihedral}} = -\frac{f_0}{n_j n_k}\frac{(\boldsymbol{n}_k\cdot\boldsymbol{r}_{ij})r_{jk}^2}{n_j^2}\,\boldsymbol{r}_{ij}\times\boldsymbol{r}_{jk}$$

$$\begin{aligned}
\boldsymbol{F}_j^{\mathrm{dihedral}} = \frac{f_0}{n_j n_k}\Bigg\{ &\frac{(\boldsymbol{n}_k\cdot\boldsymbol{r}_{ij})r_{jk}^2}{n_j^2}\,\boldsymbol{r}_{ij}\times\boldsymbol{r}_{jk}\\
&+\frac{(\boldsymbol{n}_k\cdot\boldsymbol{r}_{ij})(\boldsymbol{r}_{ij}\cdot\boldsymbol{r}_{jk})}{n_j^2}\,\boldsymbol{r}_{ij}\times\boldsymbol{r}_{jk}\\
&+\frac{(\boldsymbol{n}_j\cdot\boldsymbol{r}_{kl})(\boldsymbol{r}_{jk}\cdot\boldsymbol{r}_{kl})}{n_k^2}\,\boldsymbol{r}_{jk}\times\boldsymbol{r}_{kl}\Bigg\}
\end{aligned}$$

$$\begin{aligned}
\boldsymbol{F}_k^{\mathrm{dihedral}} = \frac{f_0}{n_j n_k}\Bigg\{ &-\frac{(\boldsymbol{n}_k\cdot\boldsymbol{r}_{ij})(\boldsymbol{r}_{ij}\cdot\boldsymbol{r}_{jk})}{n_j^2}\,\boldsymbol{r}_{ij}\times\boldsymbol{r}_{jk}\\
&-\frac{(\boldsymbol{n}_j\cdot\boldsymbol{r}_{kl})(\boldsymbol{r}_{jk}\cdot\boldsymbol{r}_{kl})}{n_k^2}\,\boldsymbol{r}_{jk}\times\boldsymbol{r}_{kl}\\
&-\frac{(\boldsymbol{n}_j\cdot\boldsymbol{r}_{kl})r_{jk}^2}{n_k^2}\,\boldsymbol{r}_{jk}\times\boldsymbol{r}_{kl}\Bigg\}
\end{aligned}$$

$$\boldsymbol{F}_l^{\mathrm{dihedral}} = \frac{f_0}{n_j n_k}\frac{(\boldsymbol{n}_j\cdot\boldsymbol{r}_{kl})r_{jk}^2}{n_k^2}\,\boldsymbol{r}_{jk}\times\boldsymbol{r}_{kl}$$

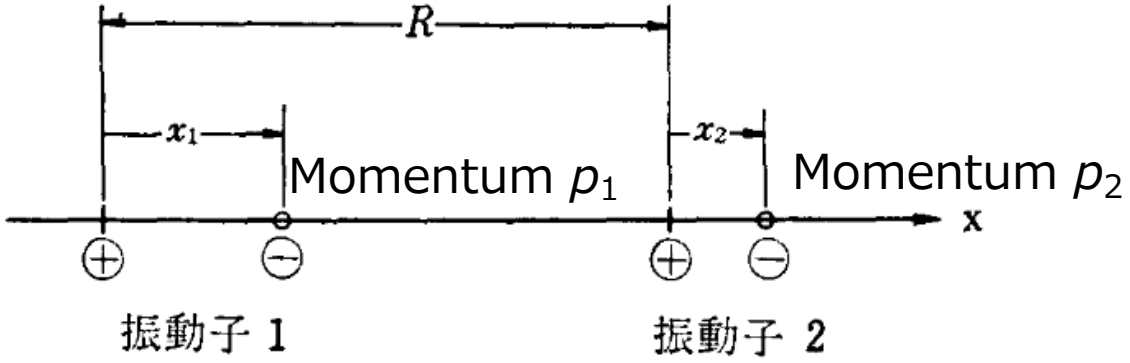
Deriving explicit forces by differentiating w.r.t. positions is often hard and tedious

Autodiff: quick to implement, faster trial-and-error cycles

Assignment

Q. Show that the attractive interaction in the Lennard-Jones potential scales as r^{-6}.

- Two atoms, each one + and one − charge (mass m); + to + distance R *(fixed)*
- Intra-atomic +/− interaction: harmonic oscillator ($1/2\ kx^2$)
- Both + charges and both − charges lie on the x-axis



1. Hamiltonian for R→∞, H_0?
2. Difference between the Hamiltonian at finite R and H_0 (= perturbation H_1)?
3. Approximate H_1 up to third order ($1/R^3$).
4. Using the transformation below, write $H = H_0 + H_1$ (to 3rd order) and show it reduces to two harmonic oscillators.

$$x_1 \equiv \frac{1}{\sqrt{2}}(x_s + x_a), \qquad x_2 \equiv \frac{1}{\sqrt{2}}(x_s - x_a)$$

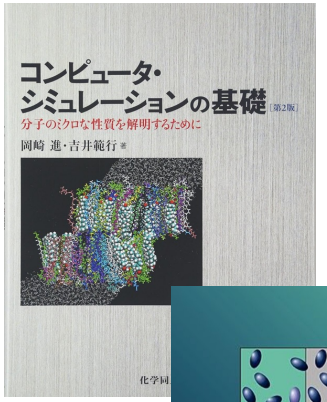
$$p_1 \equiv \frac{1}{\sqrt{2}}(p_s + p_a), \qquad p_2 \equiv \frac{1}{\sqrt{2}}(p_s - p_a)$$

5. From the zero-point energies of H and H_0, show the r^{-6} dependence.

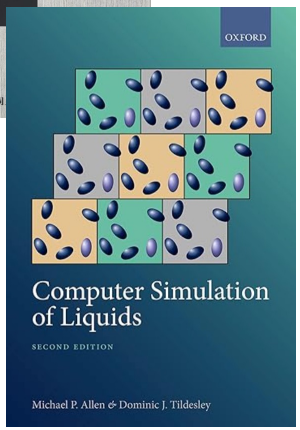
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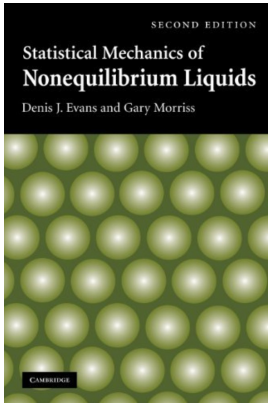


Molecular Simulation School @ IMS
– held every summer



M. P. Allen, D. J. Tildesley, “Computer Simulation of Liquids 2nd. Ed.”
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  MPRG Deep Learning Lecture Notebook
  https://github.com/machine-perception-robotics-group/MPRGDeepLearningLectureNotebook

